

STA305H5S: Project

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1 Part 1: Comparing Two Treatments

Choose fertdat (fertilizer A vs B)

1.1 Task 1: State the research question, experimental units, response, and treatment factor

Research Question: Does Fertilizer Level A have higher wheat yield than Fertilizer Level B.

Experimental units: wheat farms.

Response factor: fert - wheat yield.

Treatment factor: trt - Level of fertilizer (A or B)

1.2 Task 2: EDA: summaries and plots by treatment.

```
summary(yA); sd(yA); quantile(yA,prob=c(0.25,0.75))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  16.50   19.98   22.40   22.18   24.90   26.90

## [1] 3.857676

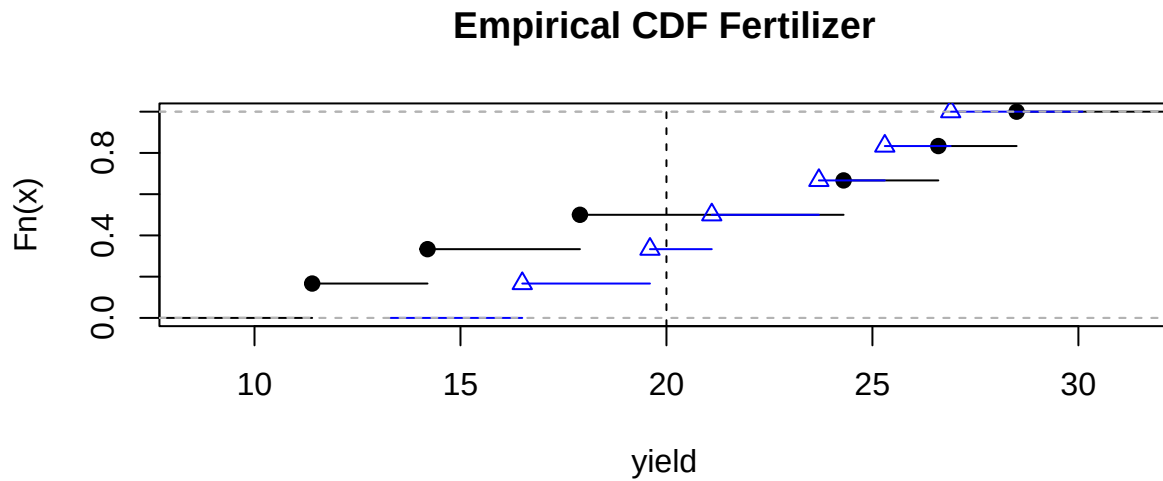
##      25%    75%
## 19.975 24.900
```

```
summary(yB); sd(yB); quantile(yB,prob=c(0.25,0.75))
```

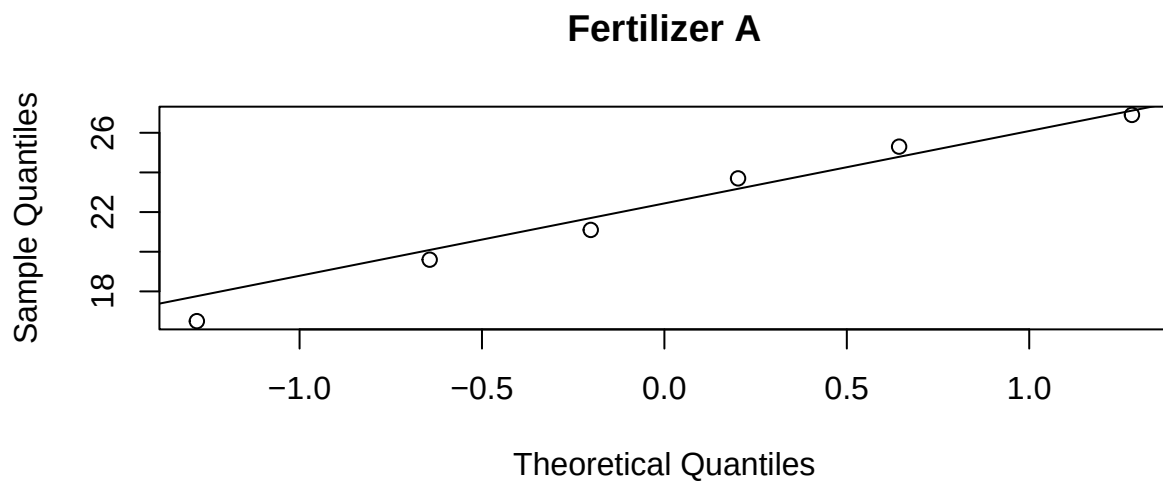
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   11.40   15.12   21.10   20.48   26.02   28.50

## [1] 6.99869

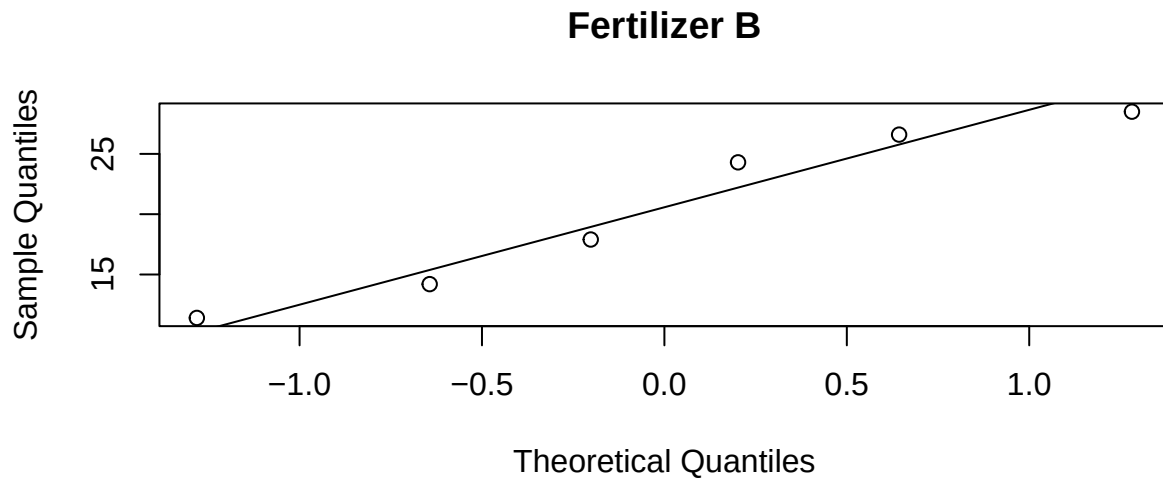
##      25%    75%
## 15.125 26.025
```



1.3 Task 3: Check normality of residuals (e.g., QQ plot, Shapiro test) and fixed variance assumptions



p-value of the Shapiro-Wilk test for yA = 0.9227 => Don't have enough evidence to say it is not normal



p-value of the Shapiro-Wilk test for $y_B = 0.5322 \Rightarrow$ Don't have enough evidence to say it is not normal.
 An F-test (var.equal) confirmed equal variances ($p = 0.217$)

1.4 Tasks 4 and 5: t.test and report the results and a 95% CI, and interpret in context.

A two-sample t-test yielded no significant difference in yields ($t(10) = 0.521$), with a p value of 0.3068, and with a 95% CI of $[-4.21, \infty]$

2 Part 2: Power & Sample Size

2.1 Task 1: Define a practically meaningful effect size

A meaningful effect size is the difference of means between y_A and $y_B = 1.7$

2.2 Task 2: Compute required sample size for 80% power at $\alpha = 0.05$ using exact method (e.g., `power.t.test()`) or Simulation-based power check.

To achieve 80% power at $\alpha = 0.05$ for the observed effect size (1.7), a sample size of 175 per group is required.
 A simulation-based check (5,000 iterations) estimated current power at only 8.22%

2.3 Task 3: Briefly discuss how sample size changes if effect size is smaller or variability is larger.

$n = \frac{2(Z_{\alpha/2} + Z_{\beta})^2 \cdot \sigma^2}{(\mu_1 - \mu_2)^2}$. This means the effect size is inversely proportional to n , that is, an increase in the effect size decreases the sample size and vice versa. The equations also shows us that if variance increases, the sample size increases, and vice versa.

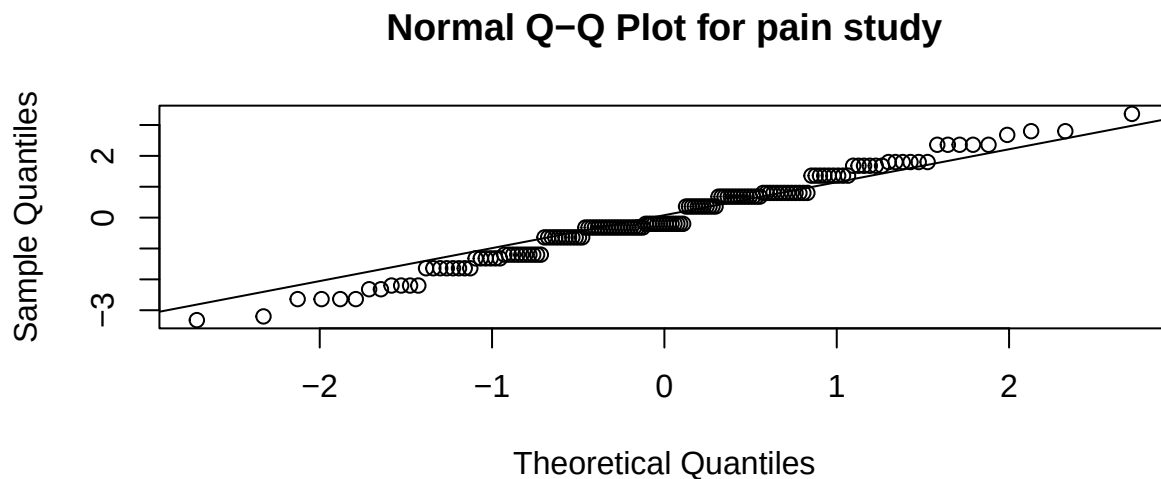
3 Part 3: More Than Two Treatments

3.1 3A: One-way ANOVA

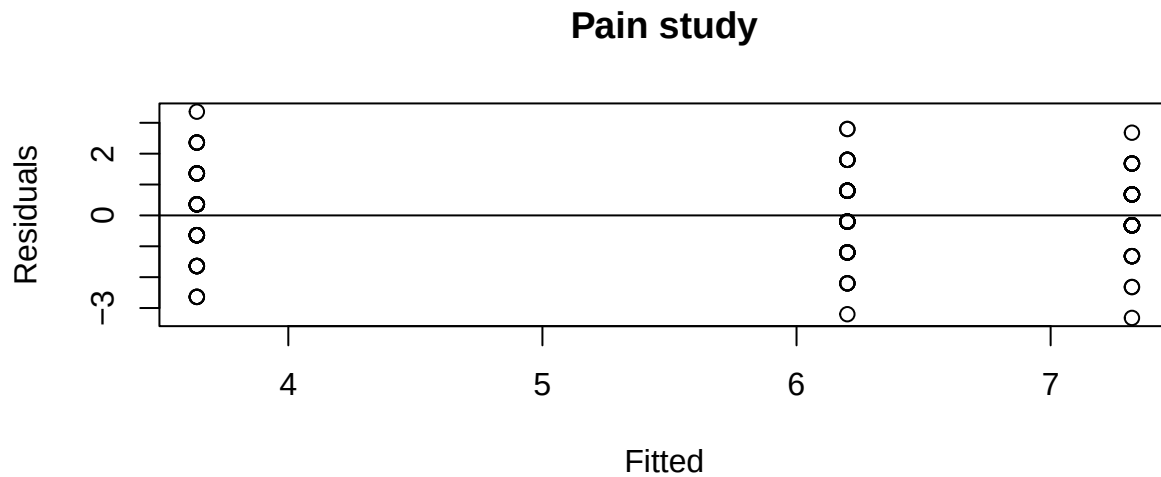
The dataset chosen is painstudy

3.1.1 Task 1: Fit one-way ANOVA; check assumptions; interpret results

The ANOVA for the pain study indicated significant differences between treatments ($F(2,147) = 98.92$) with p value $< 2e-16$



Based on the Q-Q plot, we can assume normality



Based on the plot, we can assume equality of variances.

We run a Pairwise comparisons using t tests with pooled SD.

P value adjustment method: bonferroni.

Group A vs. Group B: The p-value is 0.00015. Since $0.00015 < 0.05$, there is a statistically significant difference in pain scores between treatments A and B.

Group A vs. Group C: The p-value is $< 2e-16$. This represents an extremely high level of statistical significance, indicating that Group A and Group C are different.

Group B vs. Group C: The p-value is $< 2e-16$. This represents an extremely high level of statistical significance, indicating that Group B and Group C are different.

We also run anova:

Sum of Squares:

trt = 355.84: This is the Between-Group variation. it represents the portion of pain differences caused by the different treatments.

Residuals = 264.40: This is the Within-Group variation (or Error). It represents the “noise” or random differences between individual patients that the treatment cannot explain.

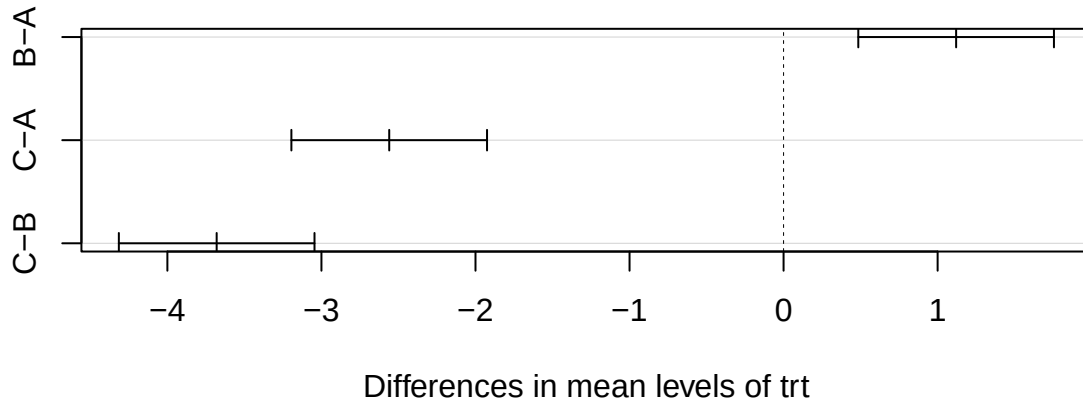
Mean Square

trt = 177.920: Calculated as $\frac{355.84}{2}$, this is the variance between treatment means.

Residuals (1.799): Calculated as $\frac{264.4}{147}$ This is the pooled variance within the groups

3.1.2 Task 2: If significant, do post-hoc comparisons (e.g., Tukey) and interpret

95% family-wise confidence level



Tukey HSD revealed significant differences between all pairs, with Treatment C showing the greatest reduction in pain compared to A and B

3.2 3B: Randomized Block Design (RBD)

The dataset used is shoedat_obs

3.2.1 Task 1: Identify the block variable and explain why blocking is appropriate.

The block variable is boy. Blocking is appropriate because each boy wore both materials, so boy-to-boy differences can be controlled for in the analysis

3.2.2 Task 2: Fit the additive model (response on treatment + block)

$$\text{Wear}_{ij} = \mu + \text{Material}_i + \text{Boy}_j + \epsilon_{ij}$$

3.2.3 Task 3: Compare conclusions vs a (hypothetical) unblocked analysis (briefly: what does blocking change?).

In the blocked model, the material effect is not statistically significant ($p = 0.220$), and the block effect for boy is also not statistically significant ($p = 0.888$). In the unblocked model, the material effect is still not statistically significant ($p = 0.136$). Therefore, the conclusion is the same in both models: there is no

significant evidence that the two materials differ in mean wear. Although material B has a slightly higher average wear than material A (11.38 versus 10.03), blocking did not change the conclusion here.

3.3 3C: Latin Square

Using dataset BR_LatSq

3.3.1 Task 1: Identify the two blocking factors (row/column) and treatment.

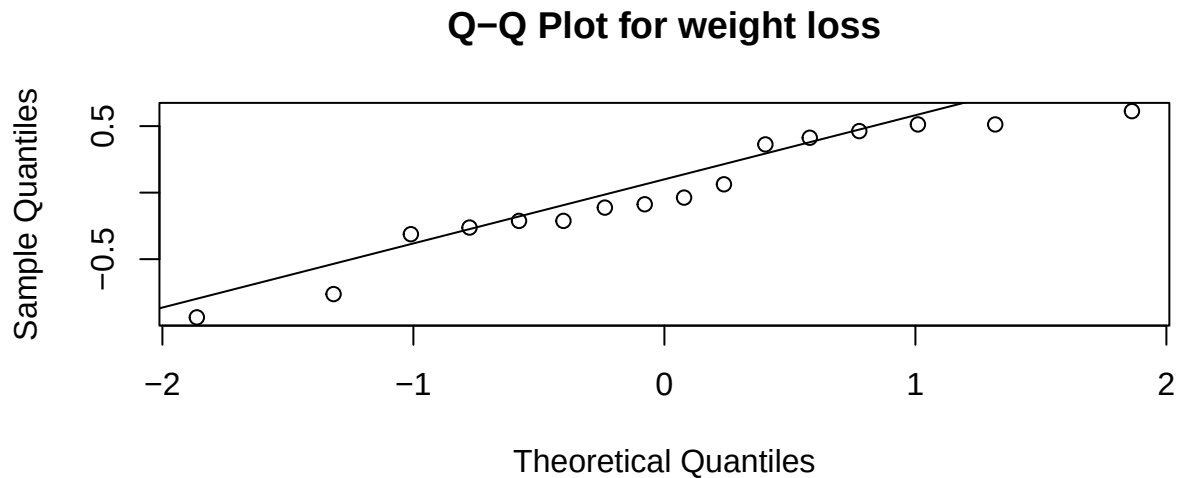
Blocking Factor 1 (Row): Dogs.

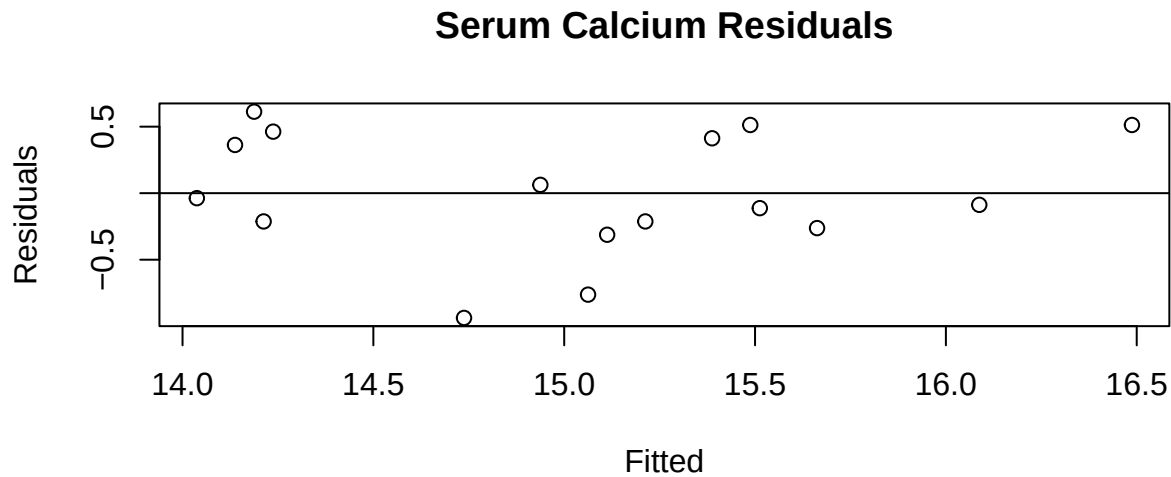
Blocking Factor 2 (Column): Weeks.

Treatment: Treat

3.3.2 Task 2: Fit response on row + col + trt

$$\hat{value} = 15.213 + 0.800Treat_B - 0.475Treat_C + 0.800Treat_D - 0.625Dog_2 - 1.075Dog_3 - 0.975Dog_4 + 0.475Week_2 + 0.075Week_3 + 0.275Week_4$$





3.3.3 Task 3: Interpret treatment effect conclusions.

The coefficients represent the difference in Mg-per cent serum calcium compared to the baseline (Treatment A):

Treat B (+0.800): This treatment increases serum calcium by 0.800 units compared to Treatment A.

Treat C (-0.475): This treatment decreases serum calcium by 0.475 units compared to Treatment A.

Treat D (+0.800): This treatment has the exact same estimated effect as Treatment B, increasing calcium by 0.800 units.

The F-statistic p-value (0.2628) and the treatment p-value (0.1162) indicate that the observed changes in serum calcium cannot be confidently attributed to the treatments. In short, the experiment does not show that the treatments have a meaningful effect.

4 Part 4: Factorial Design at Two Levels

The dataset is wtlossdat

4.1 Task 1: Identify factors, levels, and response

Factors: A - Level of food diary, B - Level of increasing activity and C - Level of home visit.

Levels: -1 and 1

Response: y - Weight loss (kg)

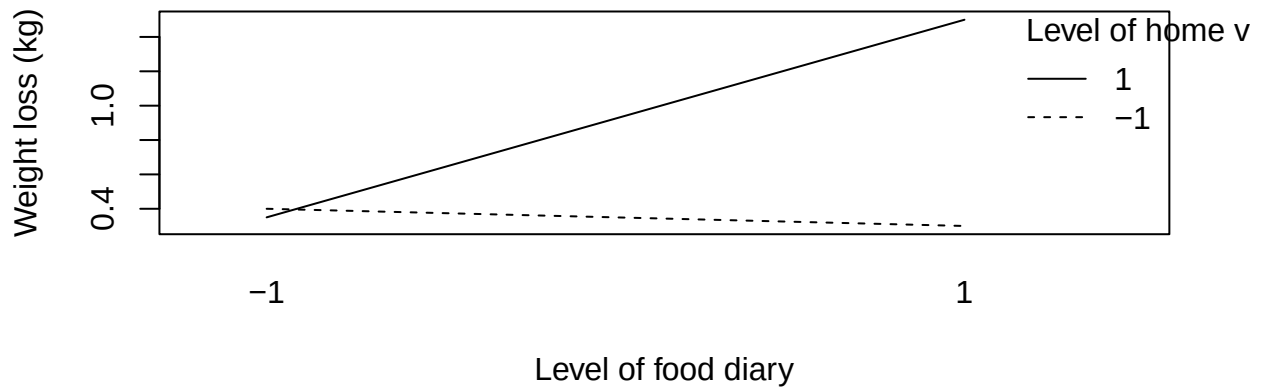
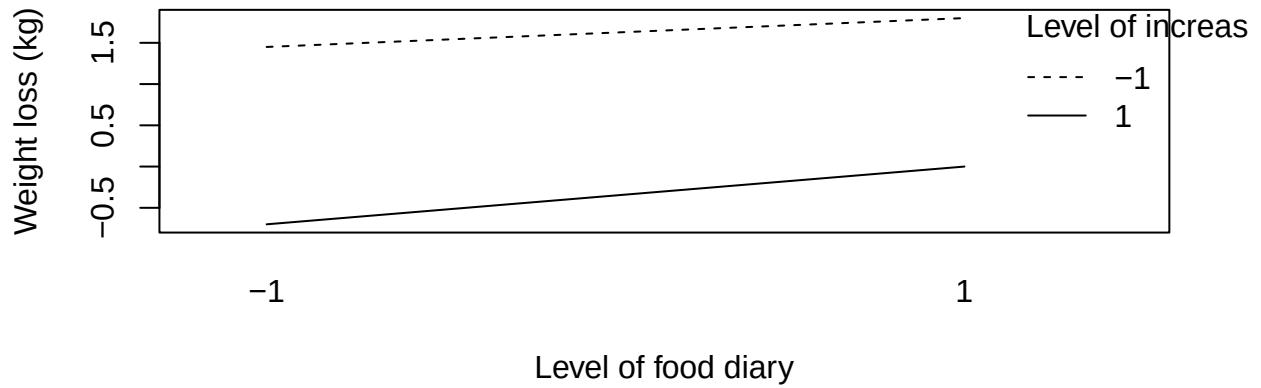
4.2 Task 2: Estimate main effects and interactions (at least two-way).

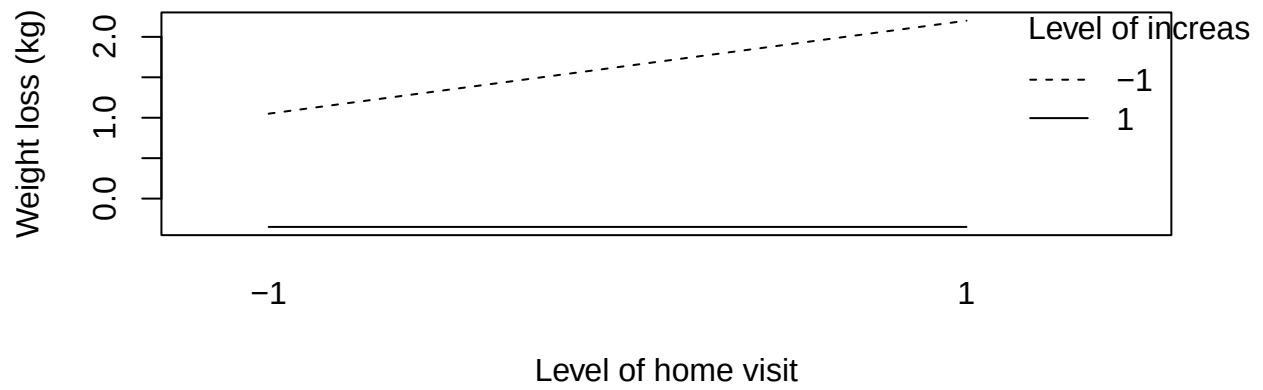
The main effects are calculated as twice the model coefficients. The estimated effect for keeping a Food Diary (A) is 0.53 kg, for Activity (B) is -1.97 kg, and for Home Visits (C) is 0.58 kg. For the two way interaction AB, the interaction effect is 0.17 and for BC it is -0.58. The strongest two-way interaction is AC (0.62 kg), suggesting a strong relation between the levels of food diary and home visits. The three way interaction effect ABC is 0.18.

4.3 Task 3: Fit an ANOVA/linear model including interactions (as appropriate).

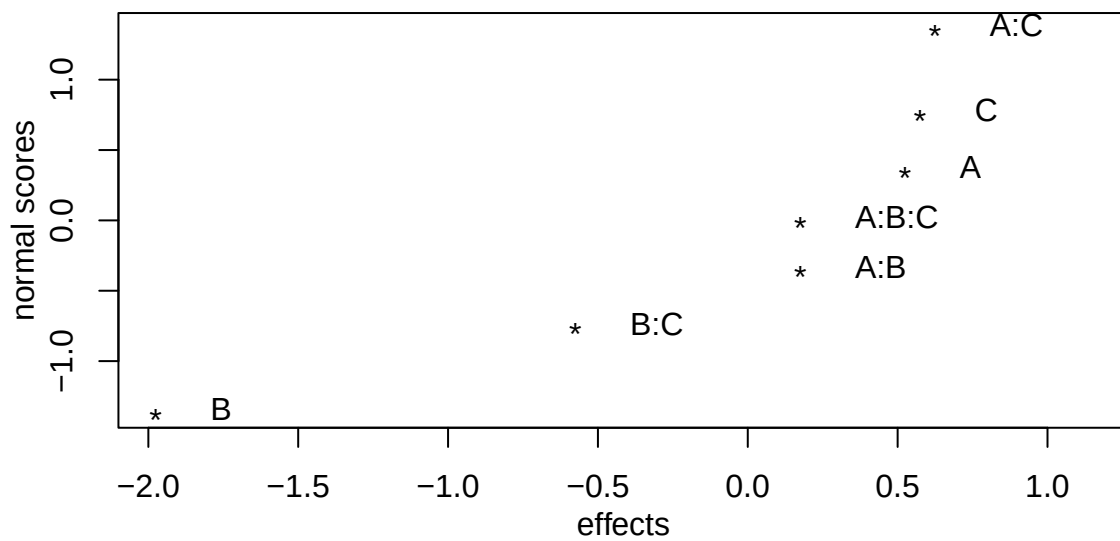
$$y = \beta_0 + \beta_A A + \beta_B B + \beta_C C + \beta_{AB} AB + \beta_{AC} AC + \beta_{BC} BC + \beta_{ABC} ABC + \epsilon$$

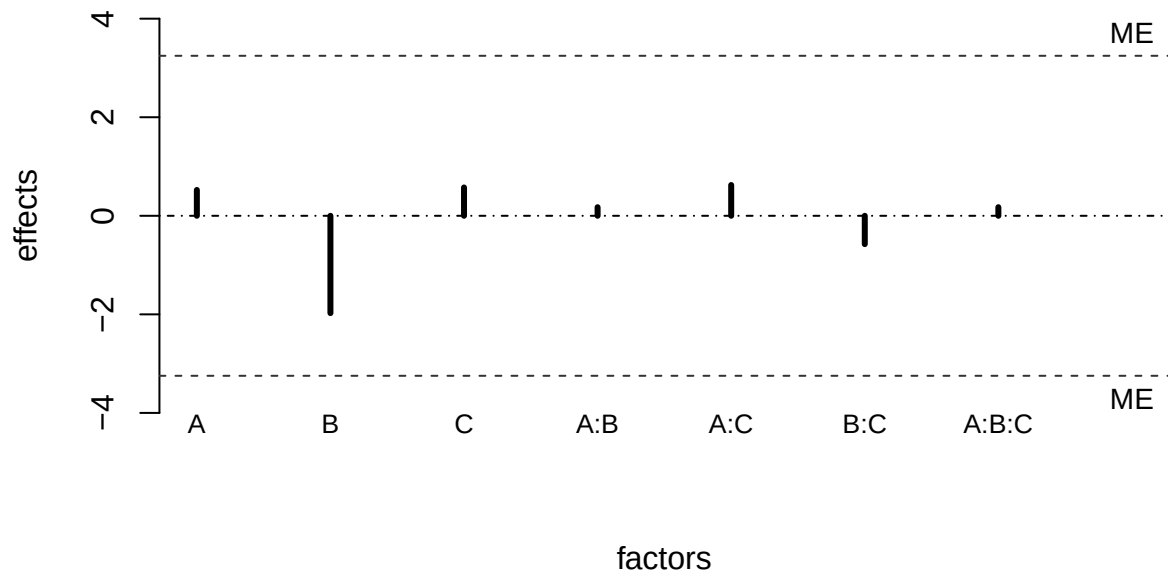
$$\hat{y} = 0.64 + 0.26A - 0.99B + 0.29C + 0.09AB + 0.31AC - 0.29BC + 0.09ABC$$

4.4 Task 4: Use plots (main effects/interaction plots) to support interpretation.



Normal plot of effects





```
##      alpha      PSE      ME      SME
## 0.050000 0.862500 3.246556 7.769665
```

4.5 Task 5: Provide a recommendation: which factor settings improve the outcome?

The factor setting that improves the outcome (maximizes weight loss) is keeping a Food Diary (Factor A: Level 1) and receiving Home Visits (Factor C: Level 1), while maintaining a Lower Level of Increasing Activity (Factor B: Level -1). This can be seen in the graphs, where B has the “highest” negative value while A, C and A:C, all have really high positive values.

5 Appendix

```

#install.packages("scidesignR")
library(scidesignR)
# See available datasets
data(package = "scidesignR")

##### PART 1 #####

View(fertdat)
??fertdat
summary(fertdat)

# RQ: Does Fertilizer Level A have higher wheat yield than Fertilizer Level B
# Experimental unit: wheat farms
# Response factor: fert - wheat yield
# Treatment factor: trt - Level of fertilizer (A or B)
fertdat$trt

# Extract yields for Fertilizer A
yA <- fertdat$fert[fertdat$trt == "A"]
# Extract yields for Fertilizer B
yB <- fertdat$fert[fertdat$trt == "B"]

##### EDA:

##### Summaries:
summary(yA); sd(yA); quantile(yA,prob=c(0.25,0.75))
summary(yB); sd(yB); quantile(yB,prob=c(0.25,0.75))

##### Plots:
wheatdat <- stack(data.frame(yA,yB))

##### Create Emperical CDF plots

```



```

plot.ecdf(yB,xlab="yield", main="Empirical CDF Fertilizer")
plot.ecdf(yA,col="blue",pch=2,add=T);abline(v=20,lty=2)

##### Create side-by-side boxplot
boxplot(values~ind,data=wheatdat)

##### Create QQPlots and Shapiro test for normality
qqnorm(yA,main = "Fertilizer A");qqline(yA)
shapiro.test(yA)
# p-value = 0.9227 => Don't have enough evidence to say it is not normal

qqnorm(yB,main = "Fertilizer B");qqline(yB)
shapiro.test(yB)
# p-value = 0.5322 => Don't have enough evidence to say it is not normal

var.test(yA, yB)
t.test(yA,yB,var.equal = TRUE,alternative = "greater") # 95 % CI
#####
##### Two-Sided Randomization p-value (Not needed because of research
##### question, but can change research question if you want to)
#tbar <- mean(res)
#pval_twosided <- sum(abs(res-tbar)>=abs(observed-tbar))/N
#round(pval_twosided,2)

# t.test(yB, yA, var.equal = TRUE)

##### PART 2 #####

power.t.test(power = 0.8, delta = mean(yA)-mean(yB),
             sd = sqrt((var(yA) + var(yB)) / 2),

```

```
sig.level = 0.05, type = "two.sample",
alternative = "two.sided")

#Simulation-based power check ####

set.seed(123)
B <- 5000
reject <- numeric(B)
for (b in 1:B) {
  xA <- rnorm(length(yA), mean = mean(yA), sd = sd(yA))
  xB <- rnorm(length(yB), mean = mean(yB), sd = sd(yB))
  reject[b] <- t.test(xA, xB, alternative = "two.sided", var.equal = TRUE)$p.value < 0.05
}
mean(reject)

##### PART 3A #####

View(painstudy)
?painstudy

pain = painstudy$pain
trt = as.factor(painstudy$trt)
aov.pain <- aov(pain~trt,data=painstudy) #anova(lm(pain~trt))
summary(aov.pain)
plot(aov.pain$fitted.values,aov.pain$residuals,ylab="Residuals",
      xlab="Fitted",main="Pain Study")
abline(h=0)

boxplot(pain~trt)
```

```
##### Checking Assumption: Normality

sapply(split(pain,trt),mean)
sapply(split(pain,trt),sd)

qqnorm(aov.pain$residuals, main="Normal Q-Q Plot for pain study")
qqline(aov.pain$residuals)
anova(lm(pain~trt))
contrasts(trt) <- contr.treatment(3); contrasts(trt)

#contrasts(trt) <- contr.sum(3); contrasts(trt)

class(trt)

lm.pain <- lm(pain~trt);
summary(lm.pain)$coefficients
model.matrix(lm.pain)

##### Checking Assumption: Constant Variance

plot(aov.pain$fitted.values,aov.pain$residuals,ylab="Residuals",xlab="Fitted",
     main="Pain study")
abline(h=0)

#### Multiple Comparison

pairwise.t.test(pain,trt,p.adjust.method = "bonferroni")
#####

pairwise.t.test(pain,trt,p.adjust.method = "none")
#####

TukeyHSD(aov.pain)
plot(TukeyHSD(aov.pain))
```

```
##### PART 3B #####

data("shoedat_obs")

shoe_long <- data.frame(
  boy = factor(rep(shoedat_obs$boy, each = 2)),
  material = factor(rep(c("A", "B"), times = nrow(shoedat_obs))),
  side = factor(c(rbind(shoedat_obs$sideA, shoedat_obs$sideB))),
  wear = c(rbind(shoedat_obs$wearA, shoedat_obs$wearB))
)

shoe_long

tapply(shoe_long$wear, shoe_long$material, mean)

model_blocked <- aov(wear ~ material + boy, data = shoe_long)
summary(model_blocked)

model_unblocked <- aov(wear ~ material, data = shoe_long)
summary(model_unblocked)

##### PART 3C #####

View(BR_LatSq)
?BR_LatSq

latinsq.auto <- lm(value~Treat+as.factor(Dogs)+as.factor(Weeks),data=BR_LatSq)
anova(latinsq.auto)
summary(latinsq.auto)

#####

##### Checking the Assumptions

qqnorm(latinsq.auto$residuals,main="Q-Q Plot for weight loss")
qqline(latinsq.auto$residuals)

###

plot(latinsq.auto$fitted.values,latinsq.auto$residuals,ylab="Residuals",
      xlab="Fitted", main="Serum Calcium Residuals")
```

```

abline(h=0)

#####

##### PART 4 #####

View(wtlossdat)
?wtlossdat

##### Interaction Plots

interaction.plot(wtlossdat$A,wtlossdat$B,wtlossdat$y, type="l",
, xlab="Level of food diary", trace.label="Level of increasing activity",
ylab="Weight loss (kg)")

interaction.plot(wtlossdat$A,wtlossdat$C,wtlossdat$y, type="l",
xlab="Level of food diary", trace.label="Level of home visit",
ylab="Weight loss (kg)")

interaction.plot(wtlossdat$C,wtlossdat$B,wtlossdat$y, type="l",
xlab="Level of home visit", trace.label="Level of increasing activity",
ylab="Weight loss (kg)")

##### Linear Model for a  $2^k$  Factorial Design

fact.mod <- lm(y~A*B*C, data=wtlossdat)
round(summary(fact.mod)$coefficients, 2)
round(2*fact.mod$coefficients, 2)

##### Full-Normal Plots

options(warn=-1)
library(FrF2)
DanielPlot(fact.mod, autolab=F, main="Normal plot of effects")

```

```
##### Half-Normal Plots
```

```
library(FrF2)
```

```
DanielPlot(fact.mod, half=TRUE, autolab=F, main="Half-Normal plot of effects")
```

```
##### Lenth's Method
```

```
library(BsMD)
```

```
LenthPlot(fact.mod, cex.fac = 0.8)
```